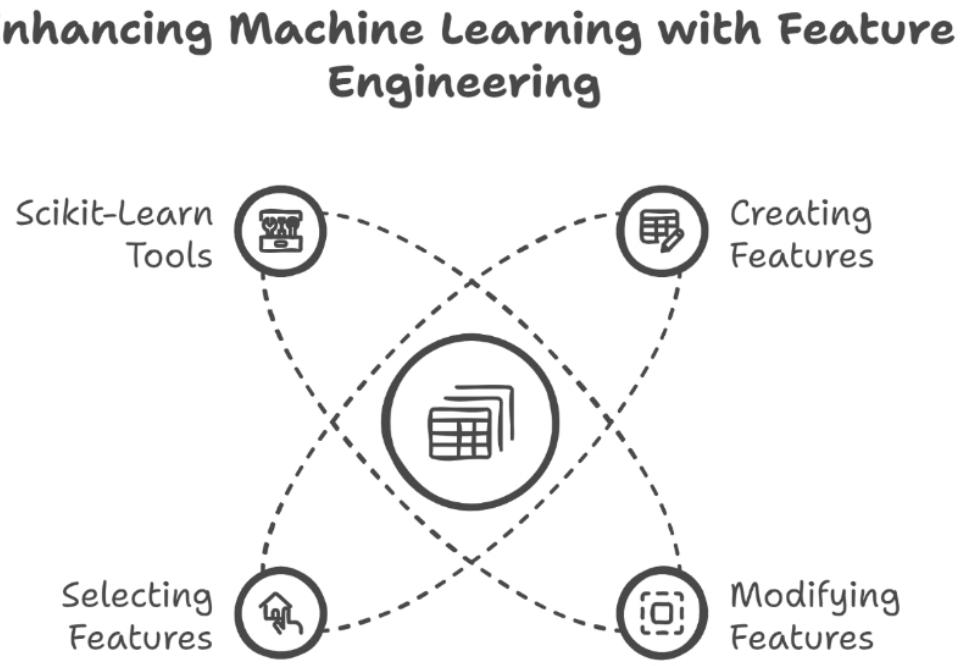


Generating Polynomial Features

Polynomial features allow models to capture non-linear relationships between variables by introducing interaction terms and higher-order terms. This is particularly useful for linear models applied to data with inherent non-linear patterns.

Implementation

Using Scikit-Learn's PolynomialFeatures class, you can generate polynomial and interaction features: from sklearn.preprocessing import PolynomialFeatures # Initialize PolynomialFeatures with degree 2

```
poly = PolynomialFeatures(degree=2)
data_poly = poly.fit_transform(data)
```

For example, if your dataset contains features x1 and x2, the resulting polynomial features will include x1, x2, x1^2, x2^2, and x1 * x2. These additional features can be fed into machine learning models to capture complex relationships.

Encoding Categorical Variables